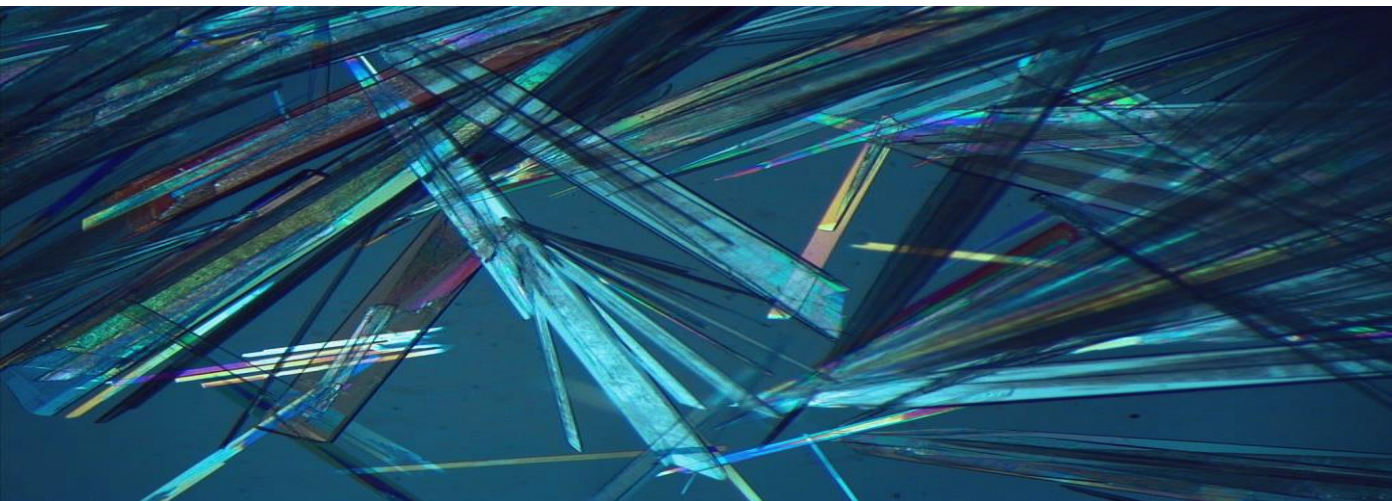




Crystal Engineering

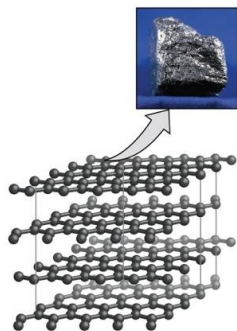
LEGO mit molekularen Bausteinen

im Spezialisierungsmodul
Advanced Materials (AdMat)

Polymorphism

„The Same and Not Quite the Same“ G. R. Desiraju

different crystal forms of the same chemical compound



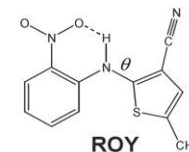
(1) R P-1
mp 106.2 °C
 $\theta = 21.7^\circ$



(2) Y P₂/c
mp 109.8 °C
 $\theta = 104.7^\circ$



(3) ON P₂/c
mp 114.8 °C
 $\theta = 52.6^\circ$



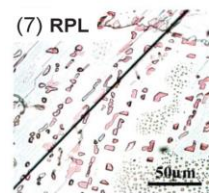
(4) OP P₂/c
mp 112.7 °C
 $\theta = 46.1^\circ$



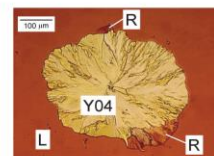
(5) YN P-1, mp 99 °C
 $\theta = 104.1^\circ$



(6) ORP Pbca
mp 97 °C, $\theta = 39.4^\circ$



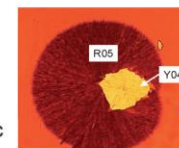
(7) RPL



(8) Y04

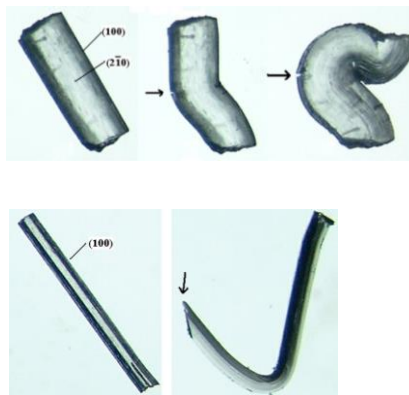
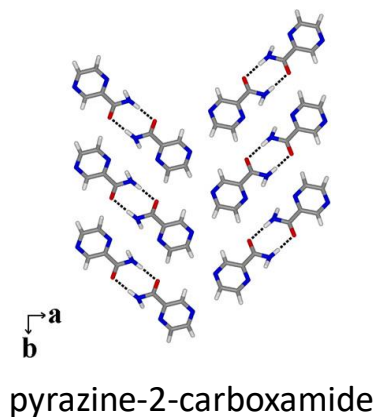


(9) YT04 P₂/c
mp 106.9 °C
 $\theta = 112.8^\circ$

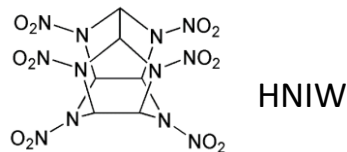
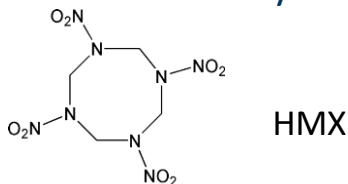


(10) R05

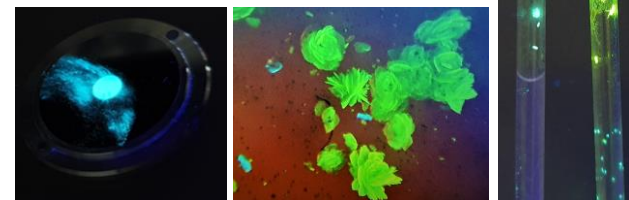
➤ Mechanical Properties



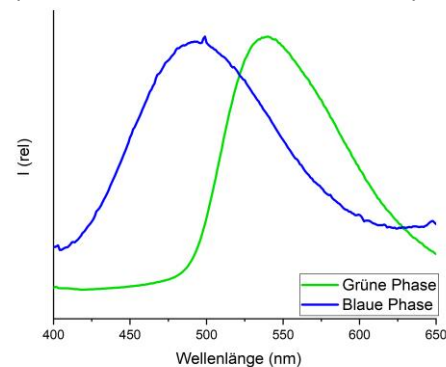
➤ Chemical Reactivity of Energetic Materials



➤ Optical properties

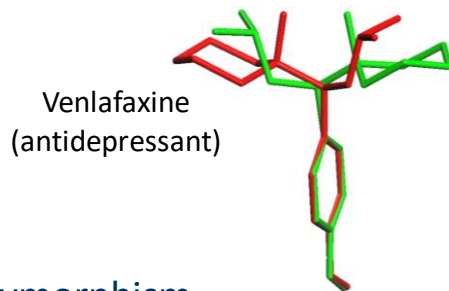


Form I Form II
of Cu(I)-HNC-complex
(from: bachelor thesis S. Hößel)

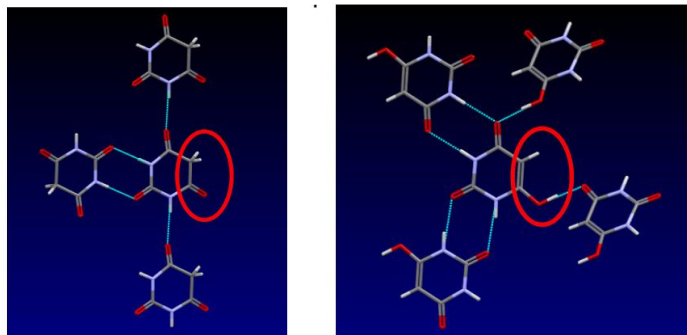


Types of Polymorphism

➤ Conformational polymorphism

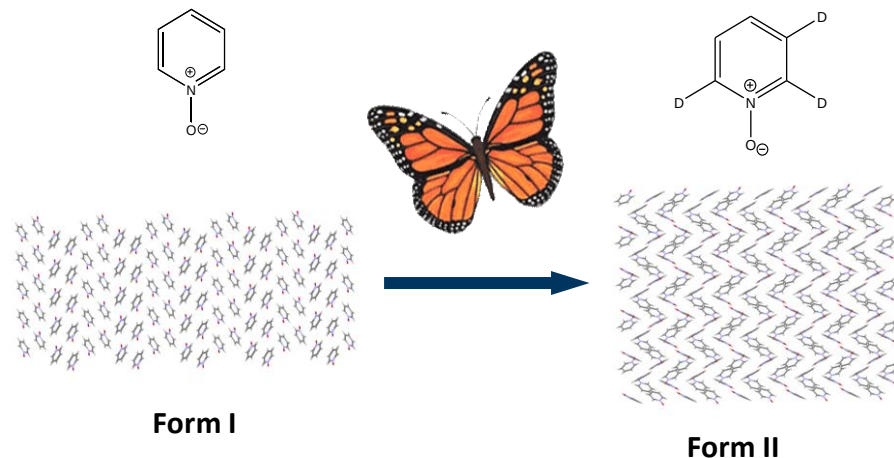


➤ Tautomeric polymorphism



Keto-Enol-TP of barbituric acid

➤ Isotopomeric polymorphism

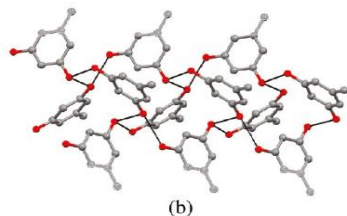
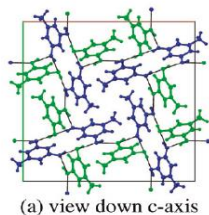


Changes in:

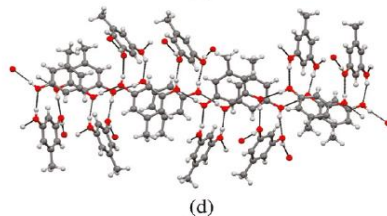
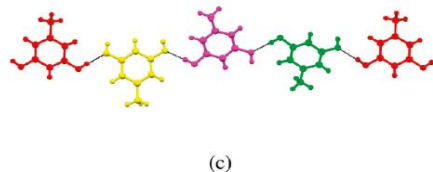
- overall packing , crystal system & space group
- intermolecular interactions

Pseudopolymorphism or Hydrates and Solvates

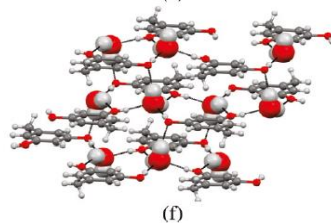
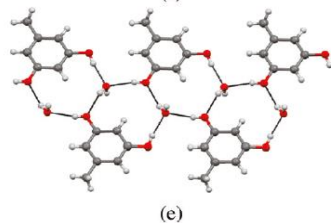
- organic molecule associated with different amounts of **solvent**
- multiple stoichiometries ($M \cdot H_2O$ / $M \cdot (H_2O)_{0.5}$ / $M \cdot (H_2O)_{1.5}$ / $M \cdot (H_2O)_2$) $M = \text{molecule}$)



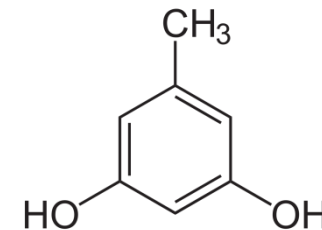
form I



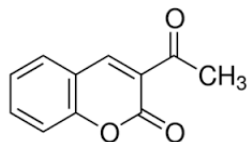
form II



monohydrate

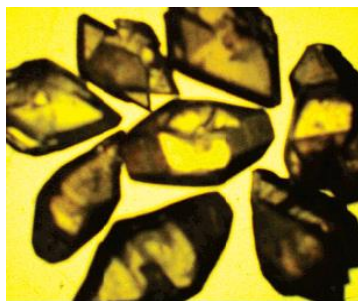


orcinol

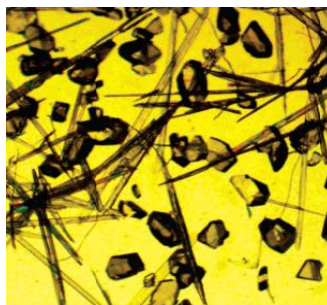


- Simultaneous appearance of different polymorphs in the same experiment under the same conditions

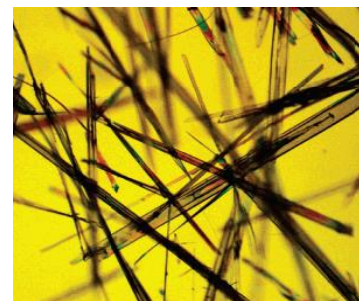
3-acetylcoumarin



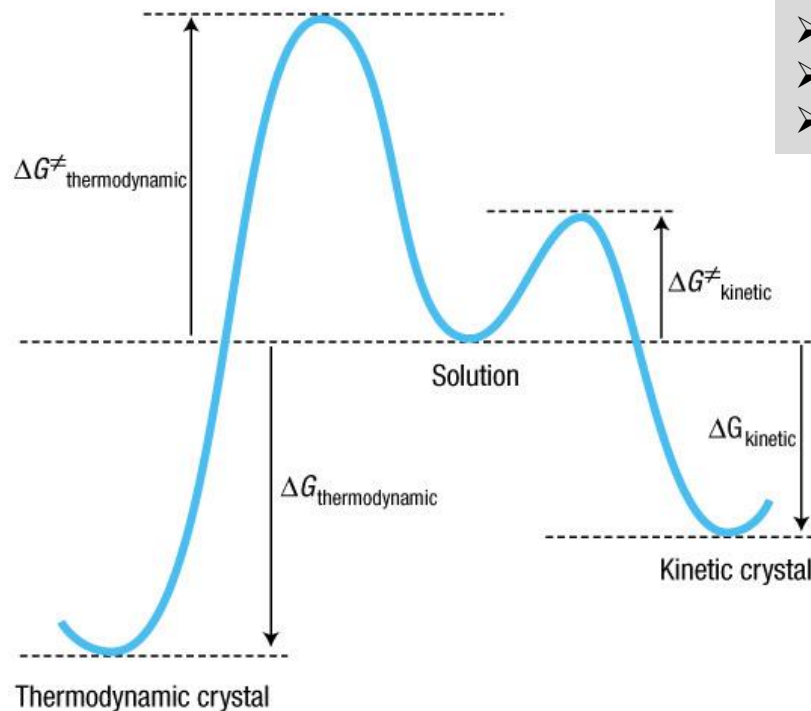
Form A
23 °C
glacial acetic acid



Form A and B
5 °C
chloroform-*n*-hexane



Form B
23 °C
chloroform-*n*-hexane



- Polymorphic systems: structures with similar lattice energies
- activation free energy are very similar
- appearance of the concomitant polymorphs possible

Ostwald's rule of stages

When leaving an unstable state, a system does not seek out the most stable state, but rather the nearest metastable state which can be reached with least loss of free energy

Polymorphs display different solid-phase physical properties

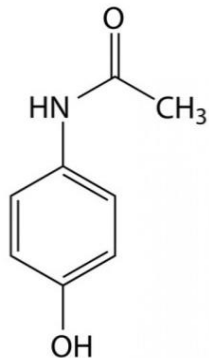
➤ polymorphism significantly affects properties of the solid form of a drug:

- solubility
- dissolution rate
- bioavailability/bioequivalence
- chemical stability ...

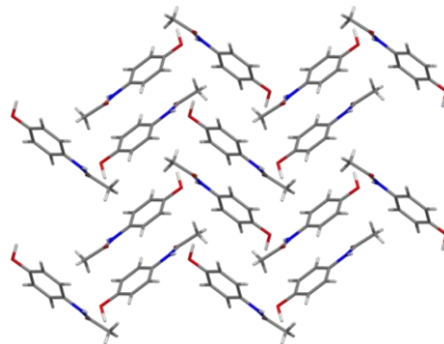
Polymorphism of Acetaminophen

Paracetamol

- an analgesic drug

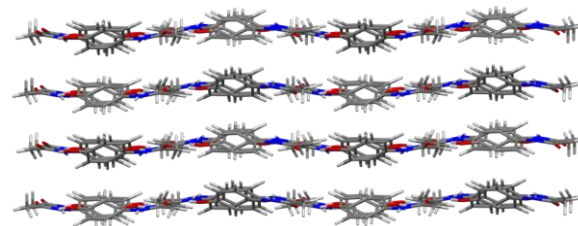


Form I (P2₁/a)



- thermodynamic polymorph
 - commercially sold
 - herringbone arrangement
 - tableting problems
 - no direct compression
 - ➔ binding agents
- costly in time and materials

Form II (Pcab)



- kinetic, metastable polymorph
 - well-developed slip planes
 - plastic deformation upon compaction
 - direct tableting
 - slightly more soluble than form I
- processing advantages & commercial benefit

Form II is better! Why is it still not on the market?

Disappearing polymorphs: Ranitidine·HCl block buster

- H2-antihistaminikum
- anti-ulcer drug
- prevent and relieve heartburn



1977: synthesis, 1st rational drug design

1978: “plate patent” of **form I** (valid till 1995)

- **Zantac**: introduced, licensed and marketed

Some time later...

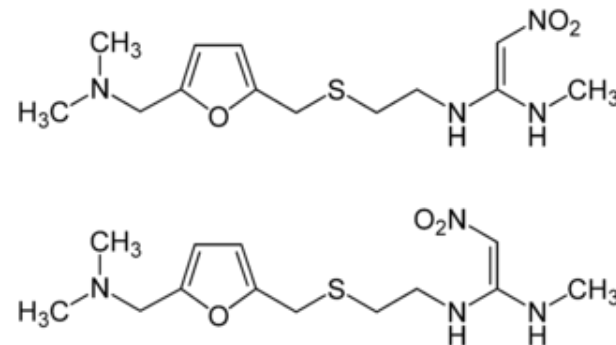
in the 13th production charge:

→ new „needle“ **form II**

Pure phase I is not producible at the Glaxo factory since then

1985: new “needle patent” of form II

- marketing



Ranitidine·HCl

a patent protection and extension block buster

Early 90-s: the sales reached 3.8 billion \$/year =
50 % of whole merit of Glaxo group



New player:

a generic pharma company **Novopharm**:

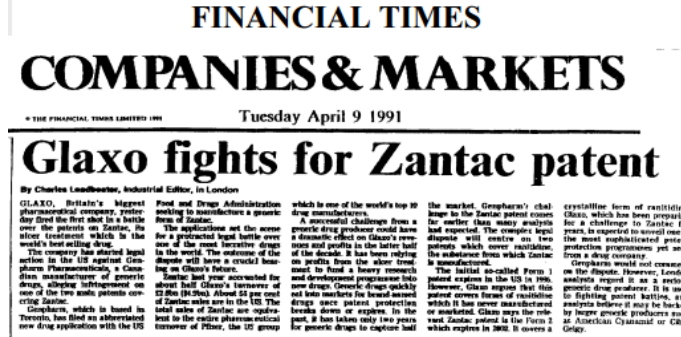
- could not reproduce the patented form I
- claim „needle patent“ to be invalid

➤ Novopharm could market form II

Glaxo sue Novopharm and defends:

- an **independent lab** can **reproduce** the protected procedure of **phase I** formation

Novopharm lost the first battle!



Ranitidine·HCl

a patent protection and extension block buster

The show goes on...

1994: **Novopharm** – new preparation method for the vanished **pure form I**

- wanted: **new patent and marketing**

Glaxo sue Novopharm again and claim:

- form I is not pure
- contains form II (protected by the „needle patent“)

Novopharm proves the purity of form I

- market a genericum Novo-Ranitidine



First major legal event!

Brought scientific field of organic polymorphism into
the glare of commercial and industrial activities

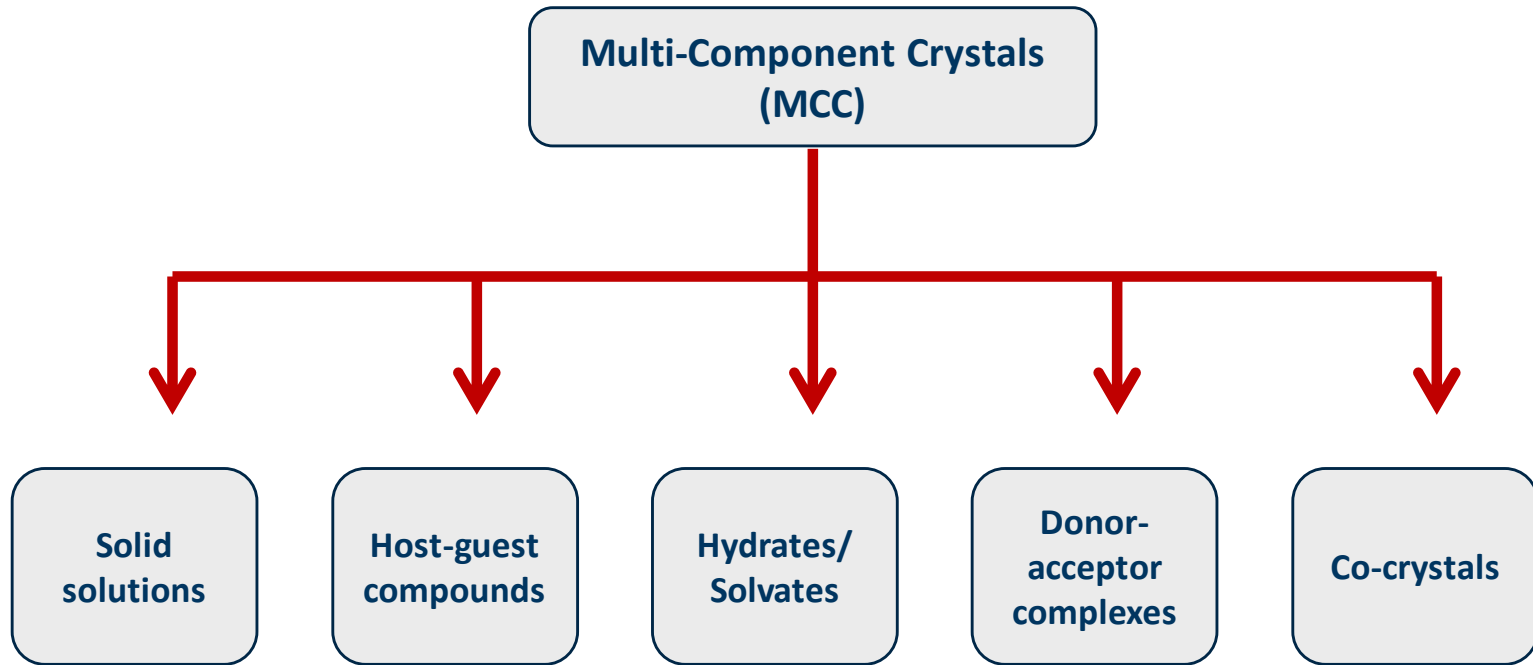
Six polymorphs:

- | | | |
|-----|--|--------------|
| I | rapid cooling of the melt | m.p. 17.3 °C |
| II | rapid cooling of the melt with 2 °C/min | m.p. 23.3 °C |
| III | crystallization at 5-10 °C | m.p. 25.5 °C |
| IV | crystallization at 16-21 °C | m.p. 27.3 °C |
| V | slow crystallization | m.p. 33.8 °C |
| | ⇒ desired edible form | |
| VI | from form V after several months at RT | m.p. 36.3 °C |
| | ⇒ thermodynamic stable form; grainy, dull surface, does not melt in the mouth | |



The controlled self-assembly of molecular species to give well defined supramolecular architectures is an important strategy for the development of functional molecular materials

**Intermolecular interactions –
useful tool for APPLICATIONS**



Multi-Component Crystals (A_mB_n)

why do some crystals contain more than one chemical component?

isolable crystals = free energy minima ΔG

In MCC: $\Delta G (A_mB_n) < \Delta G (A \text{ or } B)$

Reason for A_mB_n free energy minimisation:

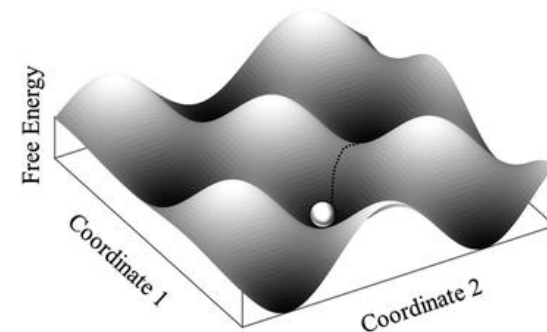
➤ **enthalpy**

- ⇒ intermolecular interactions $A \cdots B$ is more favourable than $A \cdots A$ or $B \cdots B$
- ⇒ definite composition & A:B stoichiometry (1:1, 1:2, 2:3)

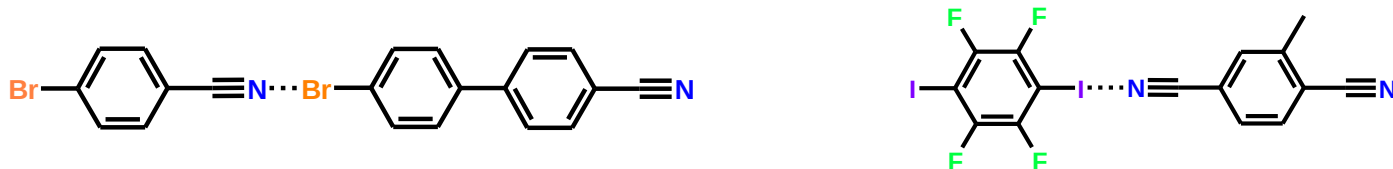
➤ **entropy**

- ⇒ molecules of the same size, shape and chemical nature
- ⇒ variable stoichiometry A_mB_n - m and n from 0 to 1

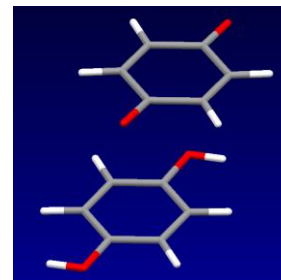
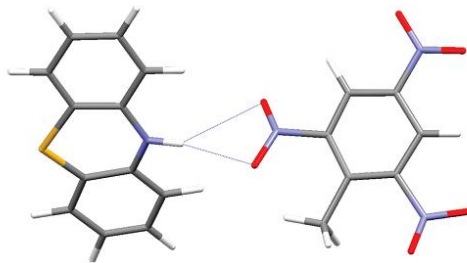
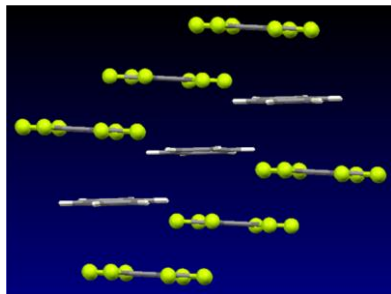
Free energy surface of a landscape



Cocrystals / co-crystals



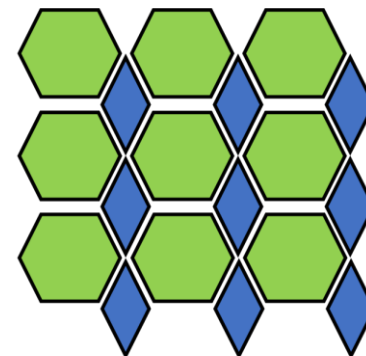
... are solids that are crystalline single phase materials composed of two or more different molecular and/or ionic compounds generally in a stoichiometric ratio.



Stoff A

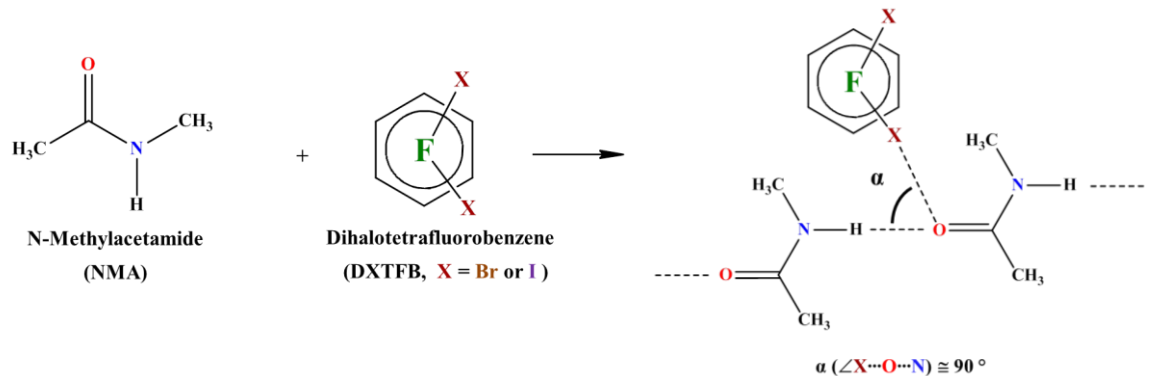


Stoff B



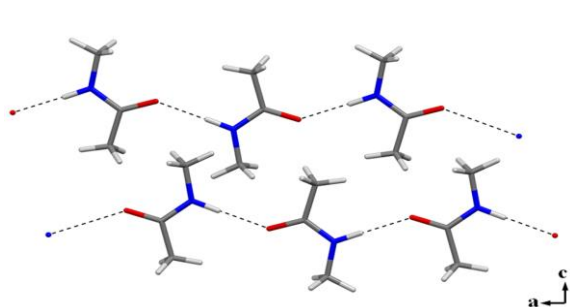
Cokristall A/B (1:1)

CE of biomimetic systems

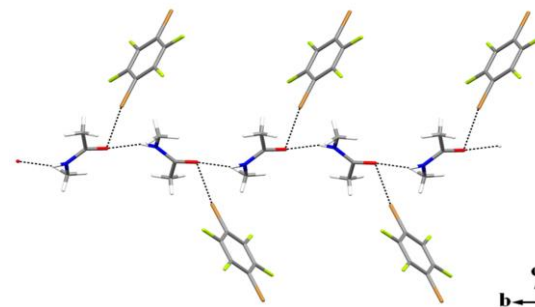


Needed:

- peptide bond model
- halogen bond acceptor
- halogen bond donor



NMA
N-H...O chains



NMA-14D^{Br}TFB
N-H...O chains & Br...O halogen bonds

Co-crystal Engineering with 2,4,6-Trinitrotoluene

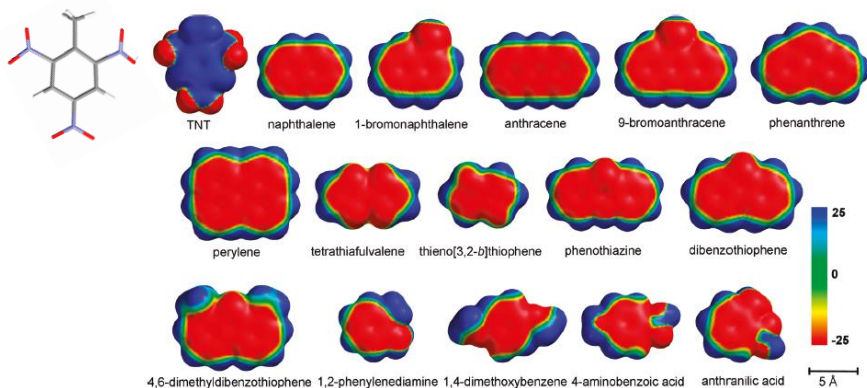


Figure 1. Electrostatic potential surfaces of TNT and the co-crystal formers calculated using the semiempirical method and the AM1 model. Red and blue surfaces represent electron rich and poor regions, respectively. Surfaces are normalized between -25 and 25 kJ/mol.

Adressed properties:

- melting point
 - packing coefficient
 - decomposition energies
- controlled explosion

Monocoordinated amine-nitro interactions
in TNT/1,2-phenylenediamine co-crystal
→ zig-zig chains generate sheets

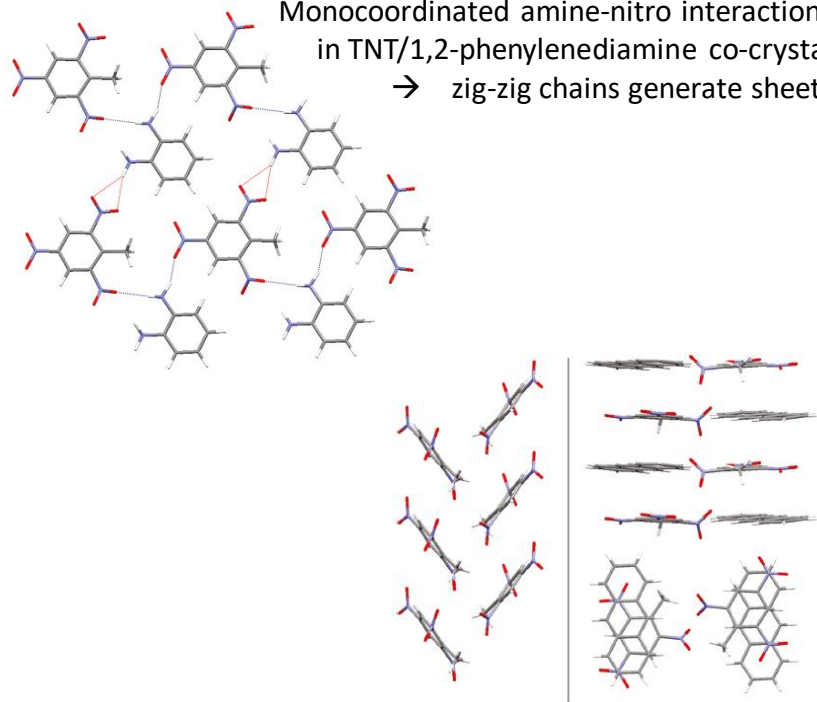
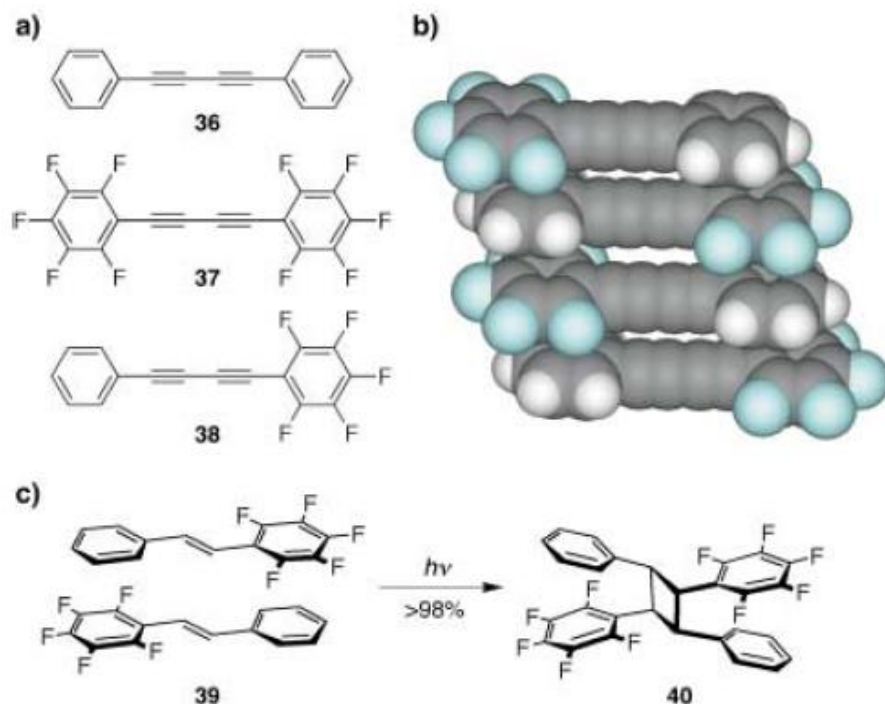


Figure 2. Comparison of the herringbone packing of monoclinic TNT (CSD code ZZZMUC08)³³ (left) and the face-to-face π -stacking of the TNT/anthracene co-crystal (right).

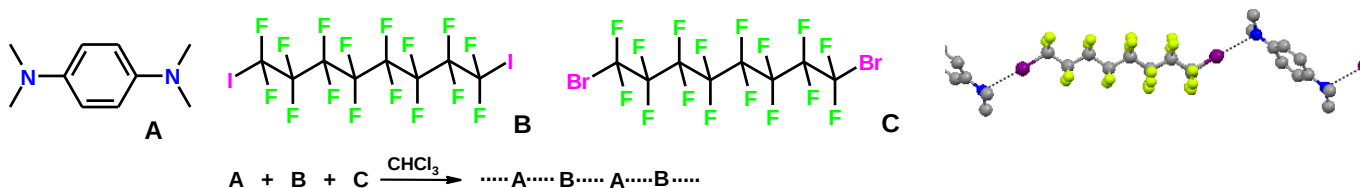
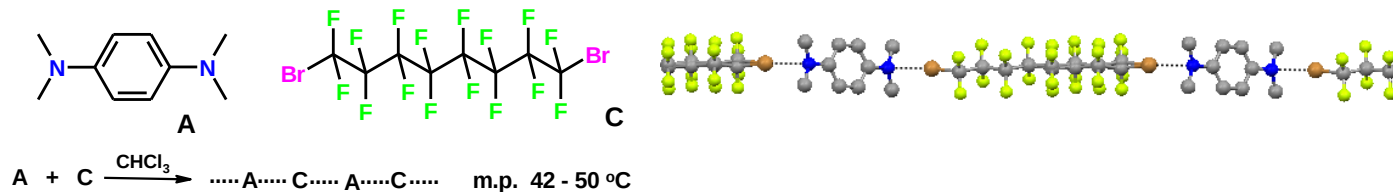
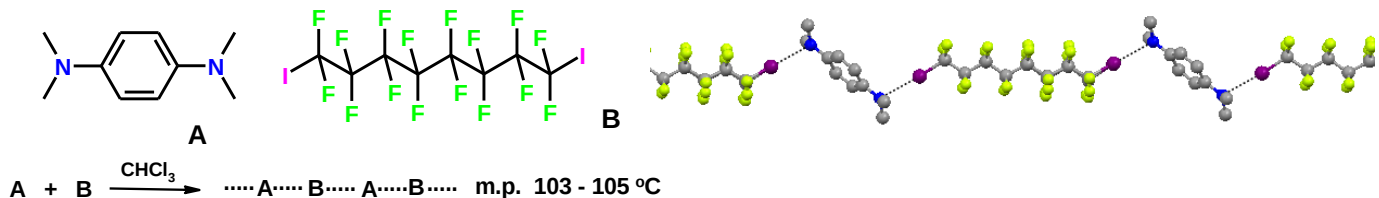
Aryl-perfluoroaryl interactions

- for controlled photoreaction in the solid state

- a) Precursors for cis-polybutadiynes.
b) The X-ray structure of **38** (CSDcode: RIHGOE).
c) The [2+2] photocycloaddition reaction of **39** proceeds with regiochemical and stereochemical control to give **40** crystals.



Selectivity in XB donors



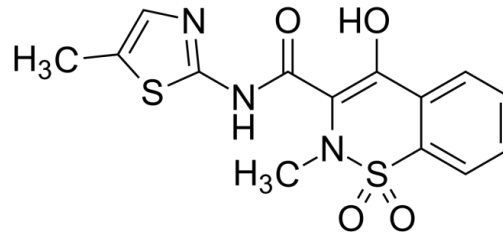
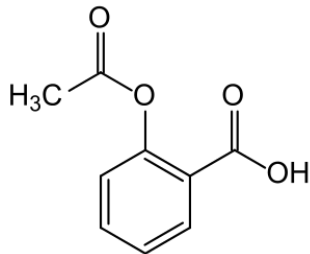
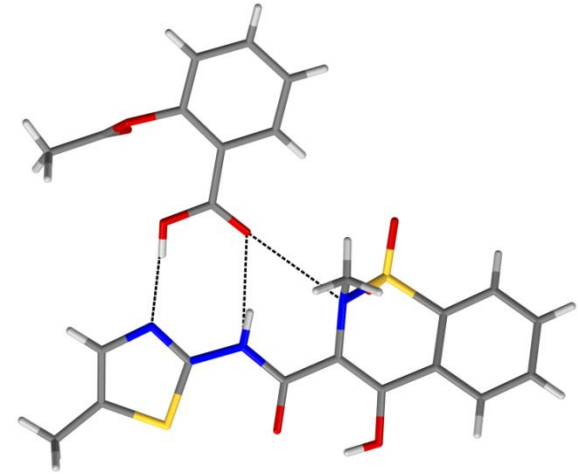
Pharmaceutical co-crystals

aspirin – meloxicam

- analgesic agent and
- non-steroidal antiinflammatory drug

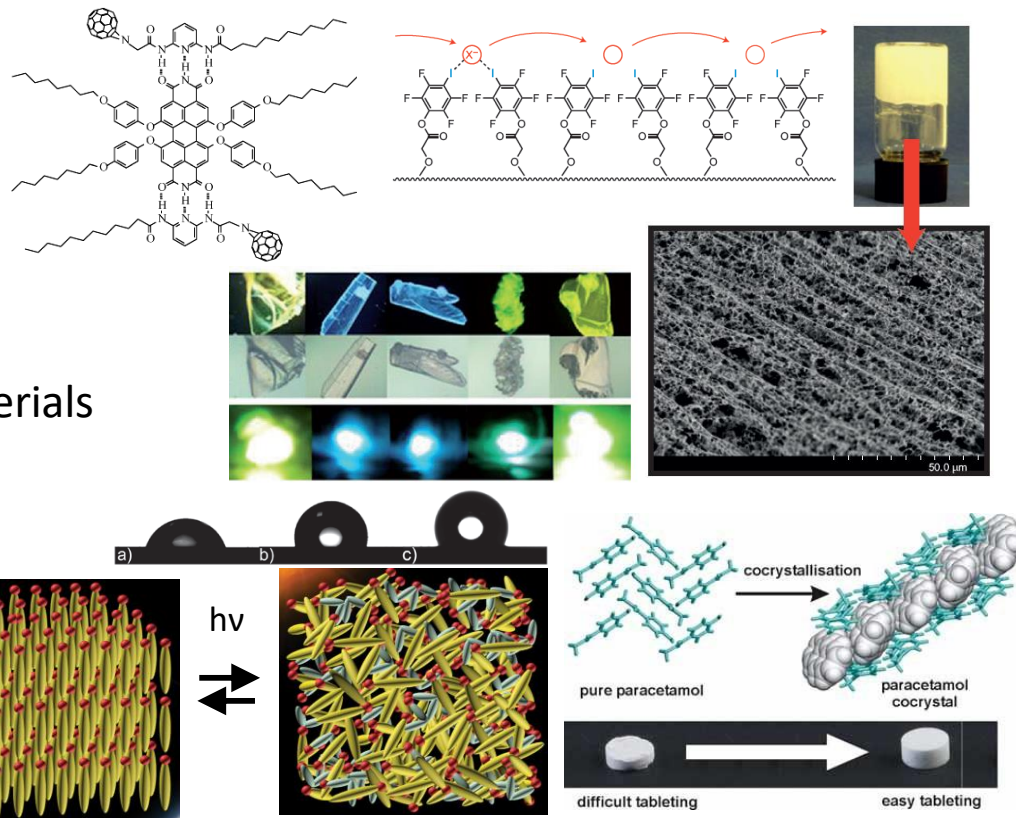
Properties:

- superior kinetic solubility
- improved pharmacokinetic properties
(decreased time for the meloxicam to reach human therapeutic concentration)
- hydrogen bonded dimer



Further potential applications:

- ⇒ optoelectronics & photovoltaics
- ⇒ small ions transporter
- ⇒ functional supramol. liquid crystals
- ⇒ low-weight gels for drug delivery
- ⇒ photoresponsive and luminescent materials
- ⇒ industrial separation
- ⇒ coatings
- ⇒ pharmaceuticals
- ⇒ ...



HOW ARE CO-CRYSTALS **DESIGNED**?

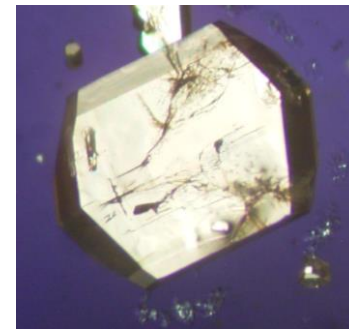
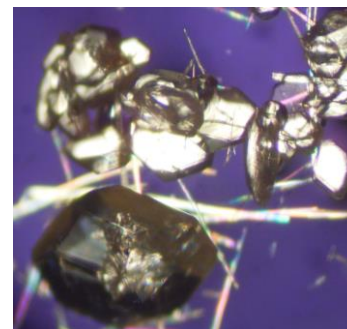
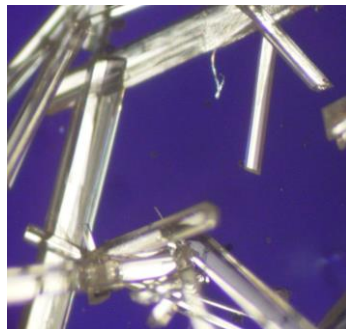
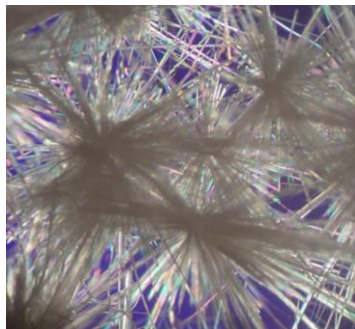
analysis of existing crystal structures via the CSD

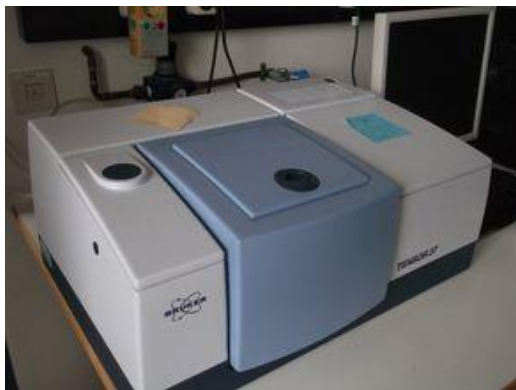
- statistical analysis of packing motifs
 - information on functional groups
 - intermolecular interactions
-
- solubility profiles

HOW ARE CO-CRYSTALS **SYNTHESISED**?

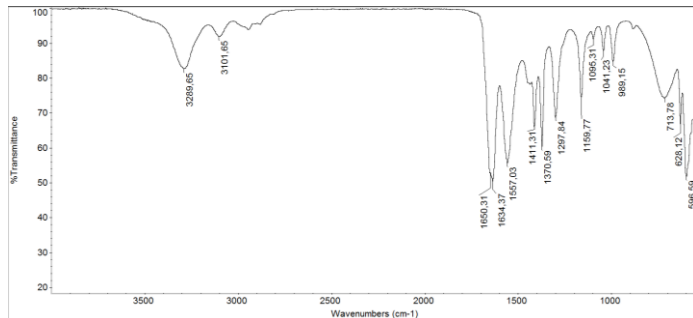
Supramolecular synthesis

- from solution (evaporations/cooling/diffusion/anti-solvent)
- mechano-chemical grinding (neat, liquid-assisted, cryogen)
- growth from the melt
- slurry

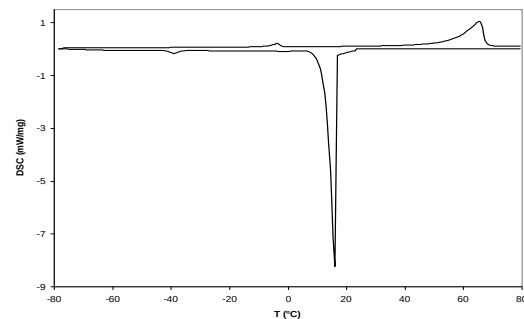




IR- and Raman-Spectroscopy



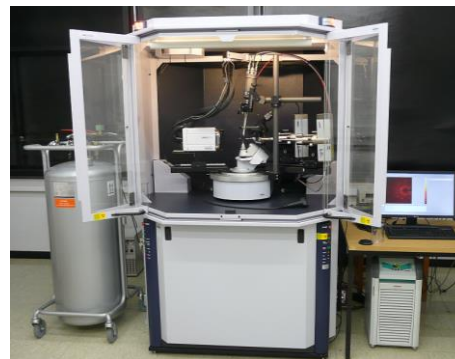
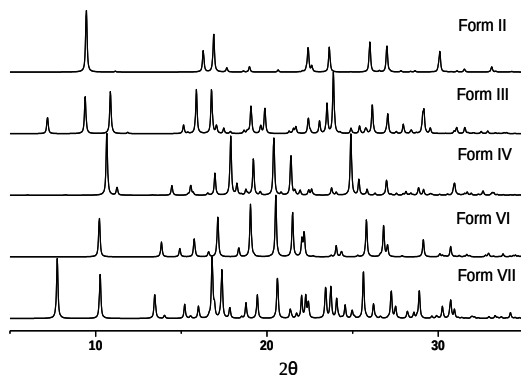
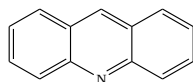
Differential scanning calorimetry (DSC) & TGA



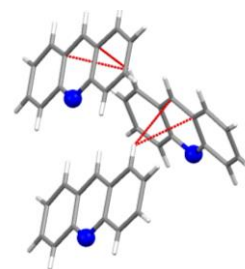
Methods of characterisation



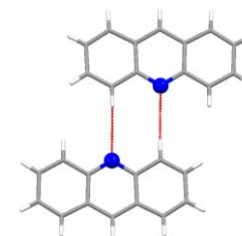
Powder X-ray diffraction (PXRD)



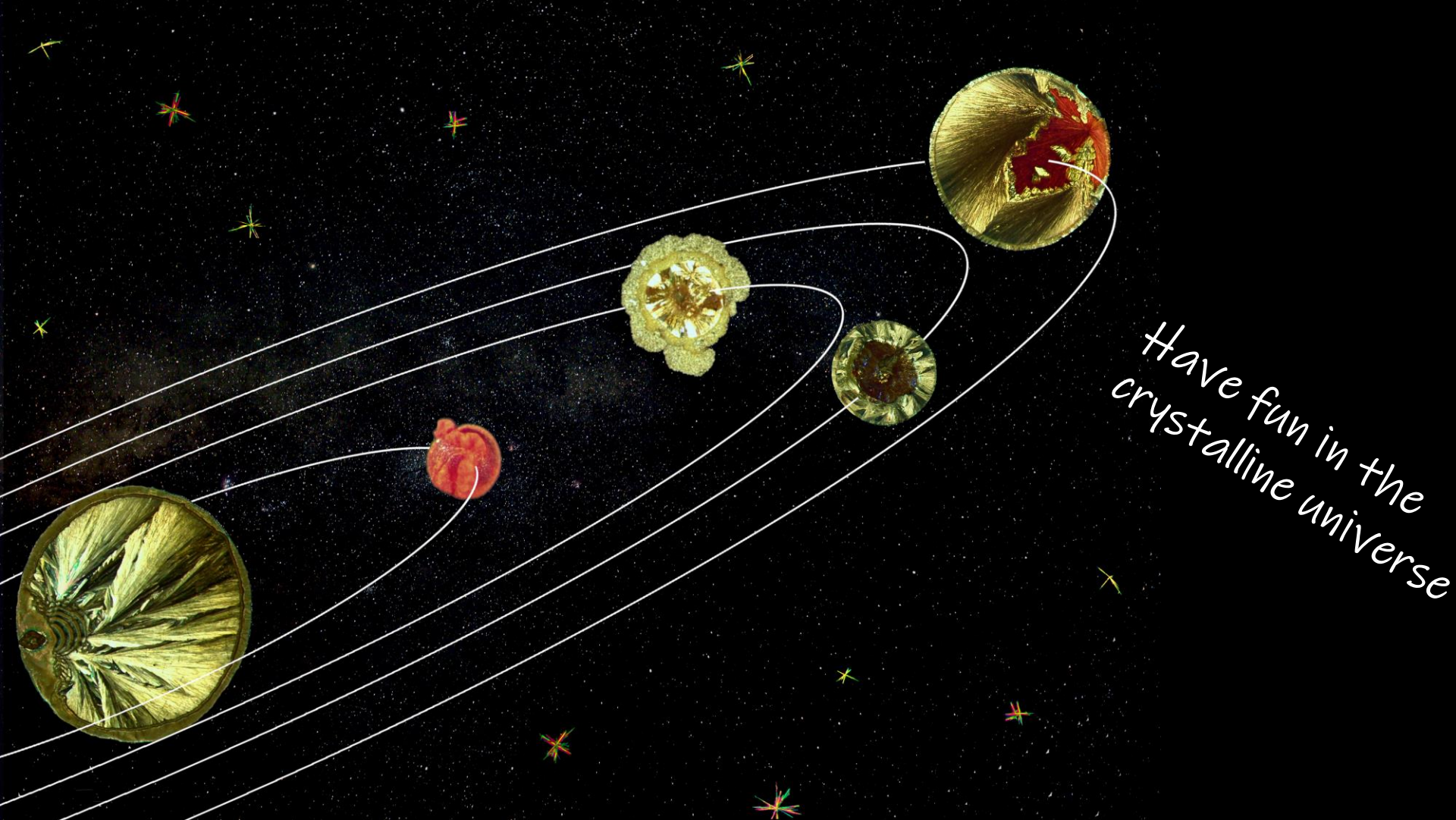
Single crystal X-ray diffraction (SC-XRD)



Form II



Form III



Have fun in the
crystalline universe